

Environment and Person Independent Activity Recognition via 80 MHz Wi-Fi Channel State Information

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Abstract—Device-free human sensing via Wi-Fi Channel State Information (CSI) is heavily hindered by spatial domain shifts and Non-Line-of-Sight (NLOS) signal attenuation. Furthermore, simultaneously solving Human Activity Recognition (HAR) and Person Identification (PI) poses an architectural conflict: HAR requires tracking macroscopic temporal kinematics, while PI relies on isolating instantaneous micro-Doppler biometric spikes. In this report, we propose a task-tailored deep learning framework to address both the HAR and the PI tasks. For HAR, we utilize a single-channel Late-Fusion Inception backbone with per-window Instance Normalization and MixStyle regularization, coupled with a BiLSTM and Temporal Attention module to model sequential movement states. Conversely, for PI, we deploy an unnormalized 4-channel Early-Fusion Inception network to strictly preserve localized, subject-specific energy profiles. Both pipelines integrate Supervised Contrastive Learning, treating 4 synchronized receiving antennas as natural multi-view positive pairs to disentangle human motion from environmental multipath noise. Experimental results across strict benchmarks demonstrate the robustness of our approach: the HAR model achieves up to 86.79% accuracy on unseen subjects, while the PI model reaches 95.10% accuracy in-domain and maintains 83.30% accuracy under severe cross-domain evaluation. Finally, we highlight that overcoming cross-domain confidence overestimation requires model checkpointing driven by Macro F1-score rather than standard cross-entropy loss.

Index Terms—Wi-Fi Sensing, Channel State Information, LSTM, Supervised Contrastive Learning, Domain Generalization.

I. INTRODUCTION

Device-free human sensing utilizing commercial Wi-Fi infrastructure has emerged as a compelling, transformative alternative to wearable sensors and vision-based systems. By analyzing how physical human movements disrupt ambient radio frequency signals, Channel State Information (CSI) enables fine-grained sensing for applications such as Human Activity Recognition (HAR) and biometric Person Identification (PI). Unlike traditional solutions, Wi-Fi sensing offers a ubiquitous, non-intrusive, and privacy-preserving medium. As bodies move within a target area, their limbs scatter wireless signals, inducing unique micro-Doppler frequency shifts and multipath interference patterns that can be captured to infer both what an individual is doing and who is doing it.

Despite its immense potential, the real-world deployment of CSI-based deep learning systems is severely hindered by the fragility of learned representations to environmental variations. This phenomenon, known as spatial Domain Shift, occurs because absolute CSI amplitude and phase are heavily dictated by static room geometries, transmitter-receiver distances, and hardware artifacts.

Consequently, standard baseline models, like basic Convolutional Neural Networks, struggle under these conditions. Furthermore, generic models fail to simultaneously capture the long-term temporal context required for activity recognition and the fine-grained biometric signatures necessary for identity verification.

To overcome these limitations, this report proposes a comprehensive and robust multi-task deep learning framework designed to decouple environmental noise from genuine human motion. The primary contribution of this work lies in demonstrating that while a unified, environment-agnostic preprocessing baseline is essential, achieving good performance across HAR and PI requires diverging, task-tailored architectural paradigms.

On one hand, HAR involves classifying continuous and macroscopic movement sequences (e.g., walking, running). We demonstrate that this task is optimally addressed through Recurrent Neural Networks equipped with Temporal Attention mechanisms such as BiLSTM, which seamlessly aggregate sequential motion states. On the other hand, Person Identification depends on isolating micro-Doppler biometric signatures. We discovered that recurrent architectures tend to over-smooth these crucial, instantaneous high-frequency spikes. Instead, PI heavily benefits from multi-scale spatial feature extraction provided by purely convolutional backbones.

To unify these task-specific models and tackle severe domain shifts, we integrate an advanced preprocessing pipeline and a multi-channel fusion strategy. Specifically, our main contributions are summarized as follows:

- **Task-Specific Architectural Design:** We evaluate multiple multi-antenna fusion schemes and reveal a clear architectural trade-off: recurrent layers (BiLSTM with Temporal Attention) are optimal for capturing continuous activity sequences (HAR), whereas pure convolutional backbones are necessary to preserve the instantaneous micro-Doppler spikes required for Person Identification.
- **Multi-View Contrastive Optimization:** We implement a Supervised Contrastive Learning (SupCon) framework that exploits the 4 synchronized receiving antennas as distinct spatial views. This forces the model to cluster inherent motion signatures while ignoring room-specific multipath interference.
- **Strict Cross-Domain Validation:** We benchmark the proposed frameworks on a comprehensive 80 MHz Wi-Fi dataset, proving their ability to overcome severe Non-Line-of-Sight (No-LOS) degradation and achieve robust Zero-Shot general-

ization in unseen rooms.

The remainder of this report is organized as follows. Section II reviews the relevant literature. Section III outlines the overall system architecture, while Section IV details the CSI data formatting and preprocessing. Section V formally describes the learning frameworks, fusion schemes, and contrastive optimization. Section VI presents the comparative experimental results and ablation studies, and Section VII concludes the paper with key observations and lessons learned.

II. RELATED WORK

The topic has been studied extensively in the recent years for activity recognition and biometric identification. Early works demonstrated the feasibility of extracting signatures and motion dynamics from CSI streams. For instance, Zeng et al. [1] showed that step patterns disrupt Wi-Fi signals in unique ways, enabling person identification in controlled indoor settings. Similarly, Wang et al. [2] developed feature extraction pipelines to map CSI speed and Doppler profiles to specific human activities.

Despite these early successes, domain dependency remains a major hurdle. Models trained in one room or on a specific set of users often experience significant accuracy drops when deployed in a new environment. To solve this, Jiang et al. [3] proposed an environment-independent framework employing adversarial domain adaptation to remove location-specific features. While adversarial training mitigates domain shift, it is notoriously prone to training instability and mode collapse.

More recently, *SHARP* [4], a state-of-the-art framework for environment-independent activity recognition using commodity IEEE 802.11 access points, was presented. By transforming CSI streams into Doppler spectrograms and applying deep neural networks, *SHARP* demonstrated remarkable accuracy and robust generalization across diverse physical settings without requiring specialized hardware.

To improve feature representation quality, Khosla et al. [5] introduced Supervised Contrastive Learning (*SupCon*), proving that pulling same-class embeddings together in the latent space yields representations superior to standard cross-entropy. Ulyanov et al. [6] showed that *Instance Normalization* can strip away sample-specific ‘style’ variations without relying on batch statistics. This property translates remarkably well to CSI data, helping us separate hardware-level gain fluctuations from actual human motion.

Building upon the baseline set by *SHARP* [4], we extend the literature in three key ways:

- 1) we show that hardware-level power fluctuations and AGC artifacts across environments are better mitigated using *Instance Normalization* [6] rather than no normalization or standard Batch Normalization;
- 2) we introduce multi-channel early and late fusion schemes paired with BiLSTM and Temporal Attention to capture fine-grained temporal and biometric signatures; and
- 3) we integrate *Supervised Contrastive Learning* [5] into the CSI pipeline. This structures the embedding space to explicitly force environment-invariant clusters, resulting in enhanced cross-domain HAR and enabling zero-shot Person Identification in completely unobserved environments.

III. PROCESSING PIPELINE

Our learning framework is designed to solve two distinct wireless sensing objectives using synchronized Wi-Fi Channel State

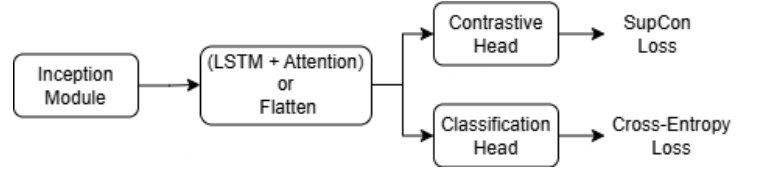


Fig. 1: Overview of the processing pipeline composed by a spatial feature backbone with domain-invariant normalization, a differential sequence modeling module for HAR, a multi-antenna supervised contrastive framework for feature disentanglement, and dedicated classification heads.

Information (CSI) represented as micro-Doppler spectrograms: Human Activity Recognition, and Person Identification. While HAR requires to suppress identity traits to isolate motion kinematics, Person ID focuses on isolating them, discarding motion trajectories. To reconcile these different requirements, we deployed two task-specific networks as represented in Fig. 1. The pipeline consists of four cohesive stages: spatial extraction, temporal modeling (for HAR), contrastive domain disentanglement and task-specific classification.

A. Multi-View Spatial Extraction & Normalization

The initial stage transforms raw multi-antenna spectrograms into robust spatial feature maps. Both networks employ a multi-branch Inception backbone using parallel kernels to capture micro-Doppler patterns across different temporal and frequency scales. To mitigate environment-induced distortions and power variations, standard Batch Normalization is replaced with Instance Normalization [6]. By computing normalization statistics independently for each spectrogram and channel, Instance Normalization neutralizes path-loss and position-dependent gain discrepancies while preserving critical velocity information. Finally, a causal MixStyle layer is integrated into the Inception module to dynamically interpolate feature statistics during training, improving generalization to unseen environments.

B. Temporal and Differential Sequence Modeling

For HAR, spatial feature maps are preserved across several temporal steps. To discriminate between motion trajectories with similar instantaneous spectral envelopes, the first-order temporal derivative ($\Delta S_t = S_t - S_{t-1}$) is computed and concatenated with primary spatial features. This joint representation is processed by Bidirectional LSTM augmented with temporal self-attention and temporal variance pooling, capturing both speed trajectories and global motion intensity. Instead, the Person ID network bypasses temporal sequence modeling, by collapsing temporal sequences into unified spatial representations. In this way the module focuses on morphological patterns.

C. Contrastive Domain Disentanglement via Multi-Antenna Views

To mitigate environmental domain shifts from room geometry and multipath noise, an auxiliary Supervised Contrastive Learning framework is integrated. Synchronized CSI traces across four receiving antennas serve as concurrent multi-view observations. A projection head maps representations to a normalized latent space where latent projections z_m, z_n ($m \neq n$) from distinct antennas act as positive pairs. Pair definitions are customized per task:

- **HAR Contrastive Space:** Positive pairs group synchronized multi-antenna views of the *same action class* across different

subjects and rooms, forcing the network to strip subject identity and isolate macroscopic kinematics.

- **Person ID Contrastive Space:** Positive pairs group views of the *same individual* across different activities and environments, compelling the network to discard action trajectories and focus on subject-specific micro-Doppler signatures.

D. Task-Specific Classification Heads

During training, the network parameters are optimized end-to-end using a multi-objective loss function that combines standard Cross-Entropy ($\mathcal{L}_{CE}$) for target classification with Supervised Contrastive Loss ($\mathcal{L}_{SupCon}$) for feature disentanglement.

Lightweight, task-specific heads apply regularized dense projections to map the disentangled representations to their respective target classes. During inference, the auxiliary contrastive projection heads are detached, and predictions are generated solely through the primary classification branches.

IV. SIGNALS AND FEATURES

To train and validate the proposed architectures across both tasks, we utilized a CSI dataset capturing human activities across multiple days, subjects, and physical environments. The feature extraction pipeline is structured into four sequential stages.

A. Measurement Setup and Input Formatting

As detailed in *SHARP* [4], the original measurements were collected using commercial IEEE 802.11ac Wi-Fi access points configured to operate over an 80 MHz bandwidth. The receiver records the complex Channel Frequency Response (CFR) across multiple subcarriers and 4 spatially distributed receiving antennas.

Rather than feeding raw complex time-series to the neural networks, the physical layer data is transformed into the time-frequency domain using a Short-Time Fourier Transform (STFT). This process generates Doppler spectrograms that effectively visualize the micro-Doppler energy shifts induced by human movements over time. The input representation is a multi-channel 2D matrix of dimensions $340 \times 100 \times 4$, representing 340 time bins (spanning an observation window of approximately two seconds), 100 Doppler frequency bins, and 4 parallel spatial receiver channels.

B. Task-Specific Pre-processing and Artifact Removal

Raw Doppler spectrograms inherently contain environmental noise and hardware artifacts. Because Human Activity Recognition (HAR) and Person Identification (PI) target different physical properties of the CSI signal, we tailor the pre-processing filtering to each specific task:

- **Human Activity Recognition (HAR):** To mitigate severe hardware shifts and power variations across different environments without destroying dynamic activity signatures, we apply *Instance Normalization* [6] across individual spectrograms. Unlike standard centering, Instance Normalization standardizes the power distribution per-sample, preserving the high-energy Doppler patterns essential for activity classification.
- **Person Identification (PI):** To isolate subtle biometric gait profiles from background environmental reflections, we perform **Temporal Mean Subtraction**. By subtracting the time-averaged mean from each Doppler bin ($x - \mu_{time}$), zero-Doppler static reflections caused by room geometry (walls,

furniture) are nullified, leaving only the fine-grained dynamic scattering generated by the moving individual.

C. Segmentation and Feature Windowing Strategy

The raw dataset consists of continuous, long-duration activity recordings. To extract discrete samples, we employ a Sliding Window Strategy over the temporal axis (340 time bins per sample). The stride parameters are carefully selected based on the task and evaluation phase to balance sample diversity and evaluation stability:

- **HAR Validation and Testing** ($S_{val, HAR} = 5$): A dense overlapping stride of 5 time bins is adopted during inference for both tasks. This guarantees high temporal resolution, eliminates temporal alignment bias, and ensures statistical consistency across continuous evaluations.
- **HAR Training Stride** ($S_{train, HAR} = 25$): A stride of 25 time bins is enforced for activity recognition training to reduce temporal redundancy between consecutive frames while maintaining sufficient dataset volume.
- **Person ID Stride** ($S_{PI} = 50$): A larger stride of 50 time bins is applied for biometric training, validation and test. Because individual gait characteristics are highly persistent over short durations, a wider stride prevents the model from overfitting to highly correlated sequential patterns.

Additionally, to improve generalization and prevent feature memorization, the obtained spectrogram windows are augmented online using SpecAugment techniques, which applies *Time Masking* (simulating a signal dropout) and *Frequency Masking* (simulating interference within a narrow frequency band).

D. Dataset Splitting

To evaluate performance under real-world domain shifts without introducing data leakage, the dataset is partitioned into distinct experimental benchmarks tailored to each task.

1) *Human Activity Recognition:* To ensure a direct and fair comparison with the literature, the HAR evaluation follows the benchmark protocol established by *SHARP* [4]:

- **Training/Validation Sets:** Constructed exclusively using data from the Environment S1, we use the first two days as training and part of the third for validation.
- **Test Sets:** Composed of the remaining part of the third recording day of Environment S1 (evaluating temporal stability in a known environment), alongside recordings from completely unobserved physical environments and unseen human subjects.

2) *Person Identification:* For the biometric evaluation, only two activities were considered: **Walking** and **Running**. Because these actions exhibit the most pronounced micro-Doppler signatures, they are inherently more distinguishable. Furthermore, to benchmark model robustness against varying degrees of domain shift, the dataset is structured into three hierarchical evaluation regimes:

- **Person 3 (In-Domain Control):** Uses a standard i.i.d. split on the S7a dataset (70% training, 15% validation, 15% testing) to establish an upper-bound baseline within a controlled environment.
- **Person 2 (Multi-Domain Robustness):** Evaluates model resilience under line-of-sight variations by training and testing across sets S3a (Line-of-Sight) and S5a (Non-Line-of-Sight).

- **Person 1 (Zero-Shot Generalization):** The most challenging setup, where models trained on controlled rooms are evaluated on entirely unobserved environments (S2a, S6a, S4b) and unseen subjects to evaluate domain-invariant biometric embeddings.

V. LEARNING FRAMEWORK

Recognizing that Human Activity Recognition (HAR) and Person Identification (PI) rely on conflicting physical phenomena, we designed two distinct architectural paradigms, both optimized via a combination of Cross-Entropy and Supervised Contrastive Learning.

A. Task-Tailored Backbone Architectures

The core feature extraction relies on a custom Inception Module, which processes the input matrices by applying parallel convolutional layers with varying receptive fields (1×1 , 2×2 , and 4×4). This guarantees the simultaneous capture of both instantaneous micro-Doppler spikes and broader temporal trends. However, the fusion strategies strictly diverge based on the task. A brief overview of the architecture is reported on Fig 2.

1) *Early Fusion for Person Identification:* To isolate biometric signatures, the network ingests the 4 antenna channels simultaneously. The input tensor $\mathbf{X}$ is processed directly by a 4-channel Inception block, forcing the network to immediately learn cross-antenna spatial correlations. Crucially, empirical evaluations demonstrated that applying Instance Normalization severely degraded identification accuracy. Localized energy variations inherently act as unique biometric identifiers; standardizing the signal locally washes away the subject’s distinct physical signature. Thus, the PI backbone strictly relies on unnormalized convolutional streams, yielding a 256-dimensional latent embedding $\mathbf{h}_{PI}$.

2) *Late Fusion and MixStyle Regularization for HAR:* To capture macroscopic temporal dynamics across domains, each of the 4 antenna views is processed independently as a single-channel input. For HAR, we extensively applied 2D Instance Normalization after each convolution. Unlike PI, macroscopic activity recognition benefits from equalizing the energy contrast locally within each sample, making the representation inherently invariant to hardware Automatic Gain Control variations across different rooms.

Furthermore, to explicitly enhance Zero-Shot domain generalization, we integrated a MixStyle regularization module within the HAR backbone. During the training phase, MixStyle probabilistically mixes the feature mean and variance between different instances within a mini-batch. By generating novel, synthetic feature statistics, the network is forced to learn structural motion dynamics rather than relying on domain-specific environmental styles. This operation is strictly disabled during validation and inference.

The network yields a 128-dimensional embedding $\mathbf{h}_{HAR}$. During training, the final prediction is obtained via Late Fusion, averaging the independent logits outputted by the classifier head.

B. Contrastive Optimization & Task-Specific Heads

Regardless of the backbone, the latent representation $\mathbf{h}$ is routed into two distinct operational branches: a non-linear projector head $g(\cdot)$ and a task-specific linear classifier head $f(\cdot)$ optimized via Cross-Entropy loss ($\mathcal{L}_{CE}$).

To enforce robust domain adaptation, we compute the Supervised Contrastive Loss ($\mathcal{L}_{SupCon}$) on these projections. For a batch of

multi-view samples, let $\mathcal{P}(i)$ be the set of positive indices for sample i , and $\mathcal{A}(i)$ be all other samples in the batch. The loss is formulated as:

$$\mathcal{L}_{SupCon} = \sum_{i \in \mathcal{I}} \frac{-1}{|\mathcal{P}(i)|} \sum_{p \in \mathcal{P}(i)} \log \left(\frac{\exp \left(\frac{\mathbf{z}_i^\top \mathbf{z}_p}{\tau} \right)}{\sum_{a \in \mathcal{A}(i)} \exp \left(\frac{\mathbf{z}_i^\top \mathbf{z}_a}{\tau} \right)} \right), \quad (1)$$

where $\tau = 0.2$ is the temperature hyperparameter. By treating the 4 synchronized antennas as natural positive views, Eq. (1) forces the encoder to tightly cluster structural velocity signatures, effectively discarding room-specific interference. The final objective is a weighted combination:

$$\mathcal{L}_{Total} = \mathcal{L}_{CE} + \lambda(t) \mathcal{L}_{SupCon}. \quad (2)$$

Following contrastive representation learning, the projection heads are discarded, and latent embeddings route exclusively through these specialized classification heads:

- **Human Activity Recognition Head:** Operates on the concatenated 384-dimensional embedding (combining bidirectional attention context and temporal standard deviation). The features pass through a regularization sequence consisting of Dropout, a Linear projection to 96 features, ReLU activation, Batch Normalization, a second Dropout layer, and a final linear classifier mapping to action classes.
- **Person Identification Head:** Operates on the flattened spatial maps (34,000 dimensions) routed sequentially through Dropout, a Linear layer mapping to 256 features, ReLU activation, another Dropout layer, and a final linear classifier mapping to subject classes.

C. Training Policies and Early Stopping

Both networks were optimized using the Adam optimizer with weight decay, coupled with a Cosine Annealing learning rate schedule.

However, model selection and early stopping criteria strictly diverged based on the distinct validation dynamics of the two tasks:

- **Person Identification:** During cross-domain validation, we observed a persistent phenomenon of *Confidence Overestimation*. The logarithmic Cross-Entropy heavily penalized single high-confidence misclassifications on out-of-distribution samples, causing the validation loss $\mathcal{L}_{val}$ to diverge prematurely even while the actual classification accuracy continued to improve. Consequently, standard loss-based early stopping was substituted. Model weights were instead checkpointed by maximizing the Validation Macro F1-Score, ensuring balanced performance across subjects.
- **Human Activity Recognition:** To prevent premature training termination while strictly avoiding overfitting, we implemented a smoothed early stopping mechanism. Specifically, training is halted if the moving average of the validation loss, computed over the last 5 epochs, strictly exceeds the best recorded validation loss for a patience threshold of 25 epochs.

VI. RESULTS

A. HAR Task

The Convolutional Recurrent Network was evaluated on the Human Activity Recognition (HAR) task across three distinct testing campaigns: **S1**, **S3**, and **S6**. These campaigns evaluate performance across different spatial environments and human subjects:

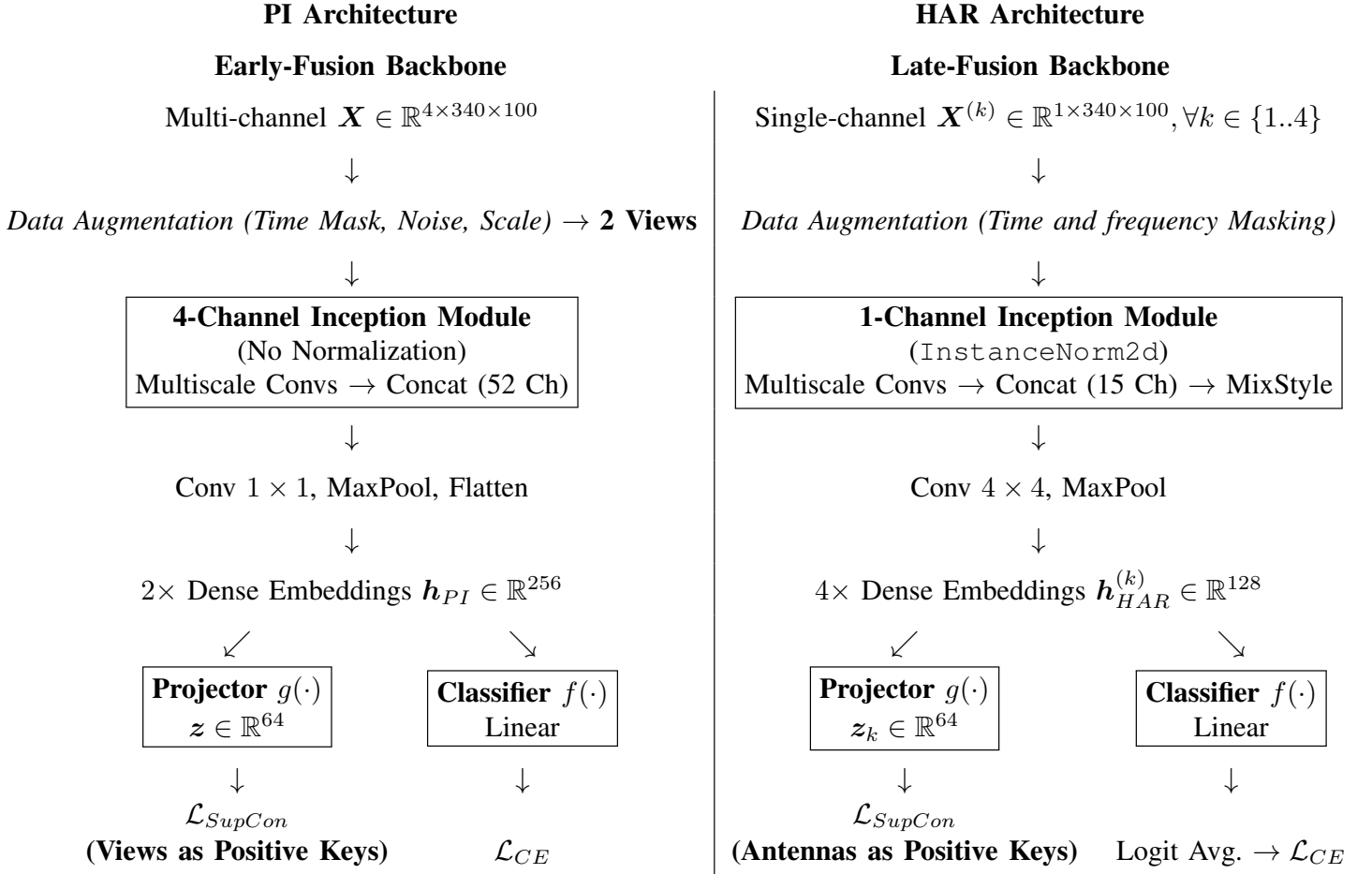


Fig. 2: Task-tailored architectural paradigms. (a) PI utilizes an Early-Fusion backbone ingesting 4 antennas simultaneously without normalization to preserve biometric energy profiles. (b) HAR utilizes independent Late-Fused channels with per-sample Instance Normalization to extract domain-invariant motion signatures.

- **S1 and S6** feature the same testing subject across different environments without obstacles (S1 in the bedroom; S6 in the living room with an altered radar/monitor setup).
- **S3** shares the same physical environment as S1 (bedroom, no obstacles) but features a different testing subject.

The model demonstrates strong overall performance, with training convergence achieved within 50 epochs. Validation accuracy stabilizes around **85.0%** after 5 epochs. The training loss decreases from 1.45 to 0.93, while validation loss settles at approximately 0.75, indicating stable convergence without significant overfitting. The fact that the validation loss is lower than the training loss is expected, as during inference for the validation process the contrastive heads are detached, and the loss is uniquely made of the cross-entropy function, naturally yielding better (i.e. lower) values.

Across all three campaigns, the model exhibits high recognition rates for static and distinct dynamic activities, while dynamic gaits (walking vs. running) present consistent boundary confusions:

- **S1 (Bedroom, Subject A):** The model achieves **100.0%** recall for Empty (E) room and **95.5%** recall for Sit (S). Jumping (J) is recognized with **85.3%** recall. Confusion occurs primarily between Walking (W) and Running (R), where Walking reaches a recall of **59.8%** (27.4% misclassified as Running) and Running achieves **77.2%** (13.5% misclassified as Walking).

- **S3 (Bedroom, Subject B):** Empty achieves **100.0%** recall. Sit achieves **73.1%** recall, with **26.9%** misclassified as Empty (E). Walking exhibits higher confusion with Running, reaching **48.3%** recall (**49.5%** misclassified as Running). Running performs strongly at **87.7%** recall (**10.8%** misclassified as Walking), and Jumping reaches **93.2%** recall (**6.7%** misclassified as Walking).
- **S6 (Living Room, Subject A):** Performance remains high and consistent with S1, yielding **100.0%** recall for Empty, **98.9%** recall for Sit, and **91.5%** recall for Jumping. Walking recall is **64.0%** (24.8% misclassified as Running), while Running achieves **80.8%** (10.7% misclassified as Walking).

TABLE 1: HAR Recall (%) and Main Confusions per Campaign

Activity	S1	S3	S6	Main Confusion
E	100.0	100.0	100.0	None
S	95.5	73.1	98.9	E (26.9% in S3)
W	59.8	48.3	64.0	R (24.8%–49.5%)
R	77.2	87.7	80.8	W (10.7%–13.5%)
J	85.3	93.2	91.5	R (14.8%), W (6.7%)

B. PI Task

For the Person Identification task across the three subjects (**P1**, **P2**, and **P3**), the model achieves an overall **Accuracy of 87.3%** and a **Macro F1-score of 90.6%**.

Validation metrics converge within the first 10 epochs, stabilizing above 0.80 accuracy and 0.85 Macro F1-score. Training loss decreases smoothly to approximately 0.80, while validation loss oscillates between 1.10 and 1.60 due to subject-specific variance across validation batches.

- **Subject P1:** Achieves a recall of **83.3%** and an F1-score of **87.8%**. Misclassifications occur exclusively with subject P2 (16.7%).
- **Subject P2:** Demonstrates solid performance with a recall of **90.2%** and an F1-score of **83.9%**, with minor confusion toward P1 (9.8%).
- **Subject P3:** Achieves perfect classification performance, with **100.0%** recall and a **100.0%** F1-score, exhibiting zero confusion with P1 or P2.

TABLE 2: Person Identification (PI) Performance per Subject

Subject	Recall	F1-Score	Misclassification Rate
P1	83.3%	87.8%	16.7% (as P2)
P2	90.2%	83.9%	9.8% (as P1)
P3	100.0%	100.0%	0.0%

TABLE 3: Quantitative Performance Comparison across Human Activity Recognition (HAR) - with generalization by day, environment and person - and Person Identification (Person ID) Tasks under In-Domain and Cross-Domain Evaluation Protocols. All results are reported as percentages (%).

(a) Human Activity Recognition (HAR)

Method / Architecture	By day		By environment		By person	
	Acc	F1	Acc	F1	Acc	F1
Baseline Inception CNN	72.43%	71.92%	70.54%	71.34%	67.56%	69.54%
CNN + LSTM	78.45%	76.92%	74.32%	73.23%	73.65%	72.20%
Our HAR Net	83.52%	83.23%	80.90%	80.78%	86.79%	86.61%

(b) Person Identification (Person ID)

Method / Architecture	In-Domain		Cross-Domain	
	Acc	F1	Acc	F1
Baseline Inception CNN	98.30%	76.90%	53.72%	61.89%
Our Person ID Net	95.10%	91.95%	83.30%	87.80%

VII. CONCLUDING REMARKS

In this paper, we developed a robust Wi-Fi micro-Doppler framework for HAR and PI, demonstrating that deep learning architectures must strictly align with underlying physical dynamics. Specifically, we observed a clear task-dependent dichotomy: while BiLSTM with Temporal Attention excelled at sequential HAR modeling, it oversmoothed the instantaneous micro-Doppler spikes necessary for PI. Instead, PI required a purely convolutional Inception backbone optimized via Supervised Contrastive Learning. Furthermore, per-window Instance Normalization proved superior to standard Batch Normalization in stripping away position-dependent hardware gain variations.

Difficulties Encountered: The primary deployment challenges stemmed from severe Non-Line-of-Sight (NLOS) signal attenuation, severe dataset class imbalance in the PI task, and strict input

dimensionality constraints. Fixing spectrograms to a 340×100 resolution severely bottlenecked the temporal horizon. Consequently, highly dynamic activities exhibited overlapping frequency bursts, making it difficult to resolve distinct stride periodicities without longer observation windows. Moreover, the severe label imbalance in PI, combined with unseen NLOS environments, initially caused high misclassifications and gradient instability during cross-entropy minimization.

Lessons Learned: These failure modes yielded two crucial engineering insights. First, synthetic spectrogram augmentations often inadvertently erase subtle kinematic features; conversely, leveraging physical multi-antenna streams as natural positive pairs for Contrastive Learning is significantly more effective. Second, spatial domain shifts combined with class imbalances induce severe *Confidence Overestimation*, rendering logarithmic cross-entropy an unreliable metric. We learned that early stopping and model checkpointing must be strictly driven by the Validation Macro F1-score to ensure balanced, true out-of-domain generalization.

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